Software Requirements Specification (SRS)

Functional and Non-Functional Requirements for the AI Application

1. System Scope and Architecture

The application is an AI-powered workspace that allows users to interact with a locally hosted language model, retrieve relevant information from a knowledge base using Retrieval-Augmented Generation (RAG), manage data sources, run AI agents, and monitor application activity.

Proposed architecture

- **Frontend:** Interactive web dashboard with nine primary pages.
- **Backend:** API service responsible for authentication, business logic, AI orchestration, retrieval, and database access.
- **LLM inference:** Ollama running locally or on a configured host.
- **Relational database:** PostgreSQL stores users, conversations, document metadata, permissions, configurations, and execution records.
- **Vector database:** Stores document embeddings and supports semantic retrieval.
- **Embedding model:** Converts document chunks and user queries into vectors.
- **RAG pipeline:** Retrieves relevant chunks, applies access-control filters, and supplies context to the LLM.
- **Optional agent layer:** Executes approved tools and multi-step workflows.

2. Requirements and Relationship Notation

- **FR:** Functional requirement — what the system must do.
- **NFR:** Non-functional requirement — how well, securely, reliably, or consistently it must operate.
- **Implements:** Identifies the page or component responsible for delivering a capability.
- **Extends:** Identifies optional or conditional behavior added to an existing workflow.
- **Depends on / Includes:** Identifies a capability that another page or workflow must use.

In formal UML use-case diagrams, `«include»` represents mandatory reused behavior and `«extend»` represents optional or conditional behavior. Page-to-page dependencies should also be documented explicitly rather than using `«extend»` for every navigation link.

1. The nine pages should cover these user needs

P1 — Authentication and Access Control

Sign-in, permissions, session security, and authorization for market-data and trading connections.

P2 — Market and Portfolio Analytics Dashboard

Market overview, interactive charts, watchlists, technical indicators, portfolio performance, risk metrics, and alerts.

P3 — AI Financial Analyst

Natural-language financial queries, comparisons, screening, calculations, explanations, and evidence-backed reports.

P4 — Financial Research and Knowledge Base

Filings, earnings reports, transcripts, research notes, news, and searchable source-backed knowledge for RAG.

P5 — Market Data and Financial Data Explorer

Price histories, fundamentals, data cleaning, queries, datasets, transformations, and exports.

P6 — Strategy, Screening, Signals and Workflow Studio

Stock screeners, strategy definitions, signal generation, rule-based alerts, and scheduled research workflows.

P7 — Backtesting, Paper Trading and Execution Monitor

Backtest results, trade-level analysis, strategy comparisons, simulated positions, and execution history.

P8 — Models, Market Data, Broker and RAG Settings

Ollama, embedding models, vector database, market-data providers, brokerage connections, data refresh, and risk configuration.

P9 — Account, Workspace and Saved Research

User preferences, saved dashboards, watchlists, queries, strategies, reports, and workspace access.

PAGE 1 — Authentication and Access Control

Purpose: Ensure that only authorized users can access the application and its protected functionality.

Functional Requirements

FR-P1-01 — User Authentication

The system shall allow authorized users to sign in using the configured authentication mechanism.

FR-P1-02 — Session Termination

The system shall provide a logout function that invalidates the active session and its associated refresh credentials, where applicable.

FR-P1-03 — Password Recovery

The system shall support secure password recovery when password-based authentication is enabled.

FR-P1-04 — User Onboarding

The system shall support administrator-controlled user creation or invitations and prevent unauthorized public registration when registration is restricted.

FR-P1-05 — Role-Based Access Control

The system shall enforce permissions according to assigned roles, such as Administrator, Operator, and Standard User.

FR-P1-06 — Protected Routes

The system shall prevent unauthenticated users from accessing protected pages and APIs.

FR-P1-07 — Permission Enforcement

The system shall check permissions on the backend for sensitive operations, including connector management, model configuration, user administration, and deletion.

FR-P1-08 — Session Expiry

The system shall expire inactive sessions according to the configured policy and require reauthentication when necessary.

FR-P1-09 — Authentication Event Recording

The system shall record relevant sign-in, sign-out, failed-login, and account-security events.

FR-P1-10 — Account Protection

The system shall support rate limiting, temporary lockout, or equivalent protections against repeated authentication failures.

FR-P1-11 — Multi-Factor Authentication: The system shall support multi-factor authentication when enabled by the deployment or account policy.

FR-P1-12 — Active Session Management: Users shall be able to inspect and terminate active sessions where supported.

FR-P1-13 — Step-Up Authentication: The system shall request additional authentication before sensitive operations, such as enabling a trade-capable broker connection or changing critical security settings, when required by policy.

FR-P1-14 — Data and Trading Permissions: The system shall distinguish read-only access to market or account data from permission to submit, modify, or cancel trading orders.

FR-P1-15 — Connection Ownership: The system shall ensure that a user cannot use another user's linked market-data credentials, portfolio, or broker account without explicit permission.

Non-Functional Requirements

NFR-P1-01 — Credential Security: Passwords shall never be stored in plaintext. Password-based authentication shall use an established, appropriately configured password-hashing algorithm.

NFR-P1-02 — Transport Security: Authentication credentials and session tokens shall be transmitted over HTTPS in deployed environments.

NFR-P1-03 — Session Security: Browser sessions shall use appropriately protected cookies or an equivalent secure session mechanism.

NFR-P1-04 — Authentication Performance: Successful sign-in requests should complete within two seconds at the 95th percentile under the agreed reference test conditions, excluding external authentication delays.

NFR-P1-05 — Information Disclosure: Authentication errors shall not reveal whether an account exists or disclose sensitive account information.

NFR-P1-06 — Accessibility: Authentication forms shall support keyboard navigation, visible focus indicators, field labels, and understandable error messages.

Page Relationships

- **Implements:** Authentication, session management, and authorization for Pages 2–9.
- **Extends:** Password recovery and additional authentication factors when those options are enabled.
- **Depends on:** PostgreSQL or the configured identity provider for user and session-related records.
- **Related pages:** Page 9 manages profile and security preferences; every protected page relies on this page's authorization rules.

PAGE 2 — Dashboard

Purpose: Provide users with a consolidated overview of application health, recent activity, and commonly used features.

Functional Requirements

FR-P2-01 — Overview Metrics

The system shall display relevant application metrics, such as the number of indexed documents, conversations, agent runs, and recent failed executions.

FR-P2-02 — Service Health

The system shall display the availability of critical dependencies, including the backend API, PostgreSQL, Ollama, the selected model, and the vector database.

FR-P2-03 — Recent Conversations

The system shall display recent conversations belonging to the authenticated user or authorized workspace.

FR-P2-04 — Recent Documents

The system shall display recently added or modified documents, their indexing status, and their processing outcomes.

FR-P2-05 — Execution Overview

The system shall display recent agent and background-job executions with their current statuses.

FR-P2-06 — Quick Actions

The dashboard shall provide shortcuts to create a conversation, upload documents, explore data sources, configure models, and inspect execution history, subject to user permissions.

FR-P2-07 — Time-Based Filtering

The system shall allow supported metrics and activity summaries to be filtered by a selected period.

FR-P2-08 — Refresh and Error Handling

The dashboard shall refresh its displayed information and indicate when a metric cannot be retrieved.

FR-P2-09 — Empty States

The system shall provide meaningful messages and next-step actions when no conversations, documents, or execution records exist.

FR-P2-10 — Market Overview: The system shall display a configurable overview of relevant market indices, asset prices, daily changes, and percentage returns.

FR-P2-11 — Multi-Market Overview: The system shall allow users to select and view supported markets and exchanges, including Indian and US equities where configured.

FR-P2-12 — Market Session Status: The system shall display whether a supported exchange is open, closed, or outside regular trading hours, using the exchange's configured calendar.

FR-P2-13 — Market Movers: The system shall identify configurable lists of top gainers, top losers, high-volume instruments, and unusually large price movements.

FR-P2-14 — Sector and Market Heatmaps: The system shall visualize price or return performance by sector, industry, or another supported classification.

FR-P2-15 — Interactive Price Charts: The system shall display interactive charts for selected instruments, including line and candlestick views where the underlying data supports them.

FR-P2-16 — Chart Timeframes: Users shall be able to select supported chart intervals and historical date ranges. The available choices shall reflect the data actually provided by the configured source.

FR-P2-17 — Chart Overlays and Indicators: The system shall allow users to add, remove, and configure supported technical indicators, including SMA, EMA, RSI, MACD, Bollinger Bands, and volume-based indicators where appropriate.

FR-P2-18 — Chart Inspection: Users shall be able to inspect values at particular timestamps and adjust the visible chart range through supported interactions such as zooming and panning.

FR-P2-19 — Instrument Comparison: The system shall allow users to compare multiple instruments over the same time range using normalized performance charts and comparable return statistics.

FR-P2-20 — Fundamental Metrics: The dashboard shall display available company fundamentals, such as market capitalization, P/E, earnings per share, dividend yield, and revenue or earnings growth, with period and source information where available.

Watchlists and portfolio analytics

- FR-P2-21 — Watchlist Management: Users shall be able to create, rename, reorder, and delete watchlists and add or remove supported instruments.
- FR-P2-22 — Watchlist Monitoring: The system shall display each watchlisted instrument's available price, price change, percentage change, volume, and other configured metrics.
- FR-P2-23 — Portfolio Summary: When a portfolio has been imported or connected, the system shall show total portfolio value, available cash where known, daily change, cumulative return, and realized and unrealized P&L where the required data exists.
- FR-P2-24 — Holdings and Positions: The system shall display portfolio holdings or positions with quantity, market value, average cost where available, portfolio weight, and unrealized P&L.
- FR-P2-25 — Benchmark Comparison: The system shall compare portfolio returns against a configurable benchmark over matching periods.
- FR-P2-26 — Contribution Analysis: The system shall identify the holdings that contributed most to portfolio gains and losses over a selected period.
- FR-P2-27 — Allocation and Exposure: The system shall summarize portfolio allocation by instrument, sector, asset class, market, and currency where the data supports those breakdowns.
- FR-P2-28 — Portfolio Risk Overview: The dashboard shall display supported portfolio risk measures, such as annualized volatility, beta, maximum drawdown, and risk-adjusted returns.
- FR-P2-29 — Correlation Analysis: The system shall visualize correlations between selected instruments or portfolio holdings and allow the analysis period to be changed.

- FR-P2-30 — Risk-Loss Visualization: The system shall display historical drawdown information and supported downside-risk measures, including VaR or expected shortfall when the required assumptions and data are available.

Research, events and personalized analytics

- FR-P2-31 — Market News Feed: The dashboard shall display relevant financial news from configured sources, with publication timestamps, source attribution, and linked instruments where available.
- FR-P2-32 — Events Calendar: The dashboard shall show relevant scheduled and reported events, such as earnings announcements, dividends, stock splits, and other available corporate actions.
- FR-P2-33 — Signals and Alerts Summary: The dashboard shall display active price alerts, triggered conditions, watchlist signals, and strategy-generated events.
- FR-P2-34 — Configurable Analytics Periods: Users shall be able to select supported periods such as one day, one week, one month, year to date, and custom date ranges.
- FR-P2-35 — Dashboard Filtering: Users shall be able to filter relevant dashboard components by instrument, watchlist, exchange, sector, portfolio, and time range where applicable.
- FR-P2-36 — Custom Dashboard Layout: Users shall be able to add, remove, resize, reorder, and save supported dashboard widgets.
- FR-P2-37 — Analytical Drill-Down: Selecting a chart, instrument, holding, event, or metric shall open the relevant detailed analysis or record.
- FR-P2-38 — Analytics Export: Users shall be able to export supported dashboard data and analytical tables into documented formats, such as CSV.
- FR-P2-39 — Data Freshness Display: Every market-price or financial metric panel shall display its applicable as-of time and indicate whether the underlying data is real-time, delayed, or end-of-day.
- FR-P2-40 — Dashboard State: The system shall preserve the user's selected market, period, layout, and other supported display preferences across sessions.

Page relationships: P2 implements market and portfolio overview. It depends on P5 for data, P8 for market-data settings, and P9 for saved layouts and user preferences. It extends into P3 for AI-driven explanations, P6 for screening and signals, and P7 for detailed strategy and execution results.

Non-Functional Requirements

NFR-P2-01 — Rendering Performance: The dashboard should render its primary content within two seconds at the 95th percentile under the reference test conditions.

NFR-P2-02 — Graceful Degradation: Failure of one optional data source shall not prevent unrelated dashboard sections from loading.

NFR-P2-03 — Data Freshness: Each dependency-health indicator and asynchronously refreshed metric shall display or enforce a defined freshness policy.

NFR-P2-04 — Data Isolation: Dashboard statistics shall include only records the current user is authorized to access.

NFR-P2-05 — Responsive Layout: The dashboard shall remain usable on supported desktop, tablet, and mobile screen sizes.

Page Relationships

- **Implements:** The overview and navigation entry point for the application.
- **Depends on:** Page 1 for access control and Pages 3, 4, and 7 for activity and status data.
- **Extends:** Quick-action navigation into other pages.
- **Related pages:** Page 9 determines user preferences; Page 7 supplies execution statistics; Page 8 supplies AI-service health information.

PAGE 3 — AI Assistant and Conversation Workspace

Purpose: Allow users to interact with the language model and receive contextual responses through ordinary inference or RAG.

Functional Requirements

FR-P3-01 — Create Conversation

The system shall allow users to start a new conversation.

FR-P3-02 — Submit Messages

The system shall accept user prompts, validate them, and send valid requests to the configured AI orchestration service.

FR-P3-03 — Stream Responses

The system shall display generated responses progressively when the configured inference interface supports streaming.

FR-P3-04 — Conversation History

The system shall save conversations and messages and allow authorized users to reopen previous conversations.

FR-P3-05 — Conversation Management

The system shall allow users to rename and delete their conversations.

FR-P3-06 — Model Selection

The system shall allow users to select from the models made available to them by the model configuration service.

FR-P3-07 — RAG Control

The system shall allow users or administrators to enable or disable retrieval-augmented generation according to the configured policy.

FR-P3-08 — Context Retrieval

When RAG is enabled, the system shall retrieve relevant, authorized knowledge-base content before constructing the LLM request.

FR-P3-09 — Source Citations

The system shall display source references for retrieved documents used in the answer, where the retrieval pipeline provides those references.

FR-P3-10 — Evidence-Aware Responses

The system shall distinguish answers supported by retrieved evidence from answers generated without retrieved evidence and indicate when relevant supporting sources cannot be found.

FR-P3-11 — Stop Generation

The system shall allow users to cancel an in-progress response when the inference service supports cancellation.

FR-P3-12 — Retry and Regenerate

The system shall allow users to retry failed requests and regenerate responses, while identifying a regenerated response as a separate generation.

FR-P3-13 — Conversation Search

The system shall allow users to search or filter their accessible conversation history.

FR-P3-14 — Copy and Export

The system shall allow users to copy responses and export conversations in at least one supported format.

FR-P3-15 — Response Feedback

The system shall allow users to submit feedback on AI responses, such as helpful or unhelpful, and associate it with the relevant response.

FR-P3-16 — Error Reporting

The system shall show understandable error messages for unavailable models, inference timeouts, invalid requests, and retrieval failures.

FR-P3-17 — Usage Information

The system shall record available request metadata, such as selected model, elapsed time, token usage when reported, and retrieval status.

FR-P3-18 — Request Context

The system shall preserve conversation context within the configured context-window and history-management limits.

FR-P3-19 — Conversation Isolation

The system shall prevent one user from retrieving another user's private conversations unless an explicit sharing permission permits it.

FR-P3-20 — File Context

If file attachment is supported, the system shall validate the attachment and make it available only through the configured, authorized processing flow.

FR-P3-21 — Financial Question Answering: The assistant shall answer questions about instrument prices, returns, performance, financial metrics, market movements, and supported portfolio information using approved data sources and calculation tools.

FR-P3-22 — Instrument Identification: The assistant shall resolve ticker and company references using the selected market or exchange and ask for clarification when an instrument cannot be identified unambiguously.

FR-P3-23 — Financial Calculations: The assistant shall request validated backend calculations for supported metrics rather than relying on language-model estimates of numerical results.

FR-P3-24 — Structured Results: The assistant shall present supported analytical outputs as tables, charts, summaries, or numerical comparisons rather than forcing every result into prose.

FR-P3-25 — Natural-Language Stock Screening: Users shall be able to describe screening conditions in natural language and receive a result set produced from supported financial data.

FR-P3-26 — Company Comparison: The assistant shall compare selected companies using available financial metrics, performance periods, and clearly identified measurement units.

FR-P3-27 — Fundamental Analysis: The assistant shall summarize available income statements, balance sheets, cash-flow information, valuation metrics, and growth measures.

FR-P3-28 — Technical Analysis: The assistant shall calculate or retrieve configured technical indicators and explain the values, observation period, and rules behind supported interpretations.

FR-P3-29 — News and Filing Analysis: The assistant shall summarize financial news, earnings reports, filings, and transcripts while providing the relevant source references.

FR-P3-30 — Event Context: The assistant shall retrieve relevant events and historical data for analysis of a market movement, distinguishing observed facts from possible explanations.

FR-P3-31 — Portfolio Analysis: Where authorized portfolio data is available, the assistant shall analyze allocation, concentration, return contributors, correlations, and configured risk measures.

FR-P3-32 — Transparent Calculations: The assistant shall identify the formula or calculation method, relevant input data, time period, and key assumptions for material financial metrics.

FR-P3-33 — Data Timestamping: Financial answers shall include or make accessible the market-data timestamp and relevant reporting period, particularly when figures may change over time.

FR-P3-34 — Saved Analysis: Users shall be able to save and reopen supported analytical conversations, outputs, and associated settings.

FR-P3-35 — Tool-Based Workflow Invocation: With authorization, the assistant shall invoke approved screening, data-query, reporting, or backtesting workflows through defined backend tools.

FR-P3-36 — Trade Action Separation: Generating an analysis, signal, or strategy shall not itself submit a real-money order. An order must use the dedicated, separately authorized trading workflow.

Non-Functional Requirements

NFR-P3-01 — Response Latency: On agreed reference hardware, the system should target a first visible response within five seconds at the 95th percentile for a warmed-up local model, excluding unusually large prompts and explicitly identified cold starts.

NFR-P3-02 — Streaming Stability: Partial output shall be displayed in order, and a failed stream shall not silently mark an incomplete answer as complete.

NFR-P3-03 — Answer Traceability: RAG-backed responses shall retain sufficient source identifiers to reproduce the source list shown to the user.

NFR-P3-04 — Model Transparency: The interface shall identify the selected model and indicate when a response is AI-generated.

NFR-P3-05 — Prompt Security: The backend shall enforce input limits and treat retrieved documents as untrusted content rather than instructions that can override the application's system policies.

NFR-P3-06 — Privacy: Sensitive prompts, credentials, and retrieved private content shall not be written to ordinary application logs without an explicitly approved and protected diagnostic mechanism.

NFR-P3-07 — Failure Isolation: A failed LLM request shall not corrupt the persisted conversation or other unrelated application records.

NFR-P3-08 — Usability: Users shall be able to distinguish pending, streaming, completed, cancelled, and failed responses.

Page Relationships

- **Implements:** The main user-facing AI interaction workflow.
- **Depends on:** Page 1 for authorization, Page 8 for model and RAG configuration, Page 4 for the knowledge base, and Page 5 for approved structured-data tools.
- **Extends:** Knowledge retrieval when RAG is enabled; attachment processing when supported; response feedback and export when requested.
- **Creates records used by:** Page 7 for execution history and Page 2 for dashboard summaries.
- **Data dependency:** Conversation and message records are stored in PostgreSQL; retrieved document chunks are sourced from the vector database and their metadata from PostgreSQL.

PAGE 4 — Knowledge Base and Document Library

Purpose: Allow authorized users to upload, organize, index, search, update, and delete documents used by the RAG pipeline.

Functional Requirements

FR-P4-01 — Upload Documents

The system shall allow authorized users to upload supported document formats, such as PDF, DOCX, TXT, Markdown, and CSV, according to the enabled parsers.

FR-P4-02 — File Validation

The system shall validate file types, sizes, and content before processing documents.

FR-P4-03 — Document Metadata

The system shall store document identifiers, titles, owners, timestamps, processing status, tags, and other configured metadata in PostgreSQL.

FR-P4-04 — Text Extraction

The system shall extract machine-readable text from supported documents and report extraction failures.

FR-P4-05 — Text Chunking

The system shall split extracted text into appropriately sized chunks using the configured chunking strategy.

FR-P4-06 — Embedding Generation

The system shall generate vector embeddings for eligible chunks using the configured embedding model.

FR-P4-07 — Vector Indexing

The system shall store embeddings in the configured vector database and associate each indexed chunk with persistent document and chunk identifiers.

FR-P4-08 — Processing Status

The system shall track document processing states, including uploaded, queued, processing, indexed, failed, and deletion pending.

FR-P4-09 — Document Search

The system shall support keyword or semantic search over authorized documents, according to the capabilities enabled.

FR-P4-10 — Document Preview

The system shall allow users to inspect document metadata and available extracted text or chunk previews according to their permissions.

FR-P4-11 — Metadata Editing

The system shall allow authorized users to edit supported document metadata without unnecessarily reprocessing unchanged document content.

FR-P4-12 — Re-Indexing

The system shall allow documents to be reprocessed when their content, embedding model, chunking policy, or index configuration changes.

FR-P4-13 — Document Deletion

The system shall remove the document's searchable vectors and associated metadata when an authorized user deletes it, subject to the defined retention and backup policy.

FR-P4-14 — Duplicate Detection

The system shall identify potential duplicate uploads using a content fingerprint or equivalent mechanism.

FR-P4-15 — Access-Control Filtering

The system shall filter search results according to document and workspace permissions before the results are used as model context.

FR-P4-16 — Processing Failure Recovery

The system shall record processing errors and allow failed documents to be retried without creating uncontrolled duplicate index entries.

- FR-P4-17 — Financial Research Sources: The system shall support importing or connecting approved financial research materials, such as company filings, earnings

reports, earnings transcripts, research notes, and relevant news documents, subject to source availability.

- FR-P4-18 — Financial Metadata Extraction: The system shall associate documents with available company names, ticker symbols, reporting periods, document types, publication dates, and source identifiers.
- FR-P4-19 — Source Provenance: The system shall retain the original source, publication date, retrieval time, and document location or URL where available.
- FR-P4-20 — Financial Table Extraction: For supported documents, the system shall extract tables and preserve table headings, row and column relationships, units, and reporting periods wherever practical.
- FR-P4-21 — Document Updates and Versions: The system shall identify revised or replacement documents and retain the relationship between document versions.
- FR-P4-22 — Hybrid Search: Users shall be able to search financial documents using keyword and semantic retrieval, with metadata filtering.
- FR-P4-23 — Research Filters: Users shall be able to filter research results by company, ticker, document type, date range, and configured source.
- FR-P4-24 — Precise Evidence Links: RAG responses shall identify the relevant document and, when available, its page, section, paragraph, or table.
- FR-P4-25 — Company and Ticker Resolution: The system shall associate alternative company names and ticker references with the appropriate entity without merging unrelated instruments.
- FR-P4-26 — Saved Research Collections: Users shall be able to group permitted documents into research collections for use in subsequent queries and analysis.
- FR-P4-27 — Research Update Detection: Where a supported data-source connector permits it, the system shall identify newly published or updated documents relevant to selected instruments or collections.

Page relationships: P4 extends P3's RAG workflow with domain-specific research. It depends on P5 for consistent instrument metadata and P8 for embedding and indexing configuration. P6 may use these collections when generating research or trading signals.

Non-Functional Requirements

NFR-P4-01 — File Safety: The system shall reject prohibited file types, enforce configurable upload limits, and use appropriate malware scanning or isolated parsing where required by the deployment threat model.

NFR-P4-02 — Index Consistency: PostgreSQL metadata and vector-index records shall be reconciled after partial failures, updates, and deletions.

NFR-P4-03 — Idempotency: Reprocessing the same document version shall not create duplicate active chunks unintentionally.

NFR-P4-04 — Processing Isolation: Long-running extraction and embedding jobs shall not block unrelated interactive API requests.

NFR-P4-05 — Search Performance: Semantic retrieval should target a 95th-percentile latency of 1.5 seconds or less on the agreed reference dataset and hardware, excluding embedding cold starts and downstream LLM generation.

NFR-P4-06 — Data Confidentiality: Unauthorized users shall not discover document titles, metadata, extracted text, or chunks through search results or direct API requests.

NFR-P4-07 — Operational Visibility: Users shall receive a clear processing status and actionable error messages rather than an indefinite spinner.

NFR-P4-08 — Data Lifecycle: Deletion and re-indexing shall follow a documented lifecycle policy that addresses queued jobs, vector records, relational records, and retained backups.

Page Relationships

- **Implements:** Document ingestion, knowledge-base management, and retrieval preparation.
- **Depends on:** Page 8 for embedding and chunking settings, Page 1 for document access permissions, and PostgreSQL plus the vector database for persistence.
- **Extends:** RAG by supplying retrieved chunks to Page 3 and knowledge-retrieval tools to Page 6.
- **Related pages:** Page 7 monitors lengthy ingestion and re-indexing operations; Page 2 displays document counts and processing status.

PAGE 5 — Data Sources and Structured Data Explorer

Purpose: Allow authorized users to manage approved structured-data sources and expose controlled data access to the assistant and agents.

Functional Requirements

FR-P5-01 — Source Registration

The system shall allow authorized administrators to register supported data sources, including approved PostgreSQL connections or file-based sources where configured.

FR-P5-02 — Connection Testing

The system shall validate connection details and report success or a meaningful failure reason without exposing credentials.

FR-P5-03 — Source Inventory

The system shall list configured sources, their types, owners, connection statuses, and last successful synchronization times.

FR-P5-04 — Schema Discovery

The system shall allow permitted users to inspect the schemas, tables, and fields exposed by approved sources.

FR-P5-05 — Access Allowlists

The system shall allow administrators to restrict the tables, views, or operations available through a source connector.

FR-P5-06 — Structured Search

The system shall provide filtering, sorting, pagination, and supported search operations over authorized structured data.

FR-P5-07 — Read-Only Querying

Where natural-language-to-SQL or equivalent querying is implemented, the default execution policy shall restrict model-generated queries to approved read-only operations.

FR-P5-08 — Query Validation

The backend shall validate generated or submitted query structures, use parameterized execution where applicable, and reject prohibited operations.

FR-P5-09 — Query Results

The system shall display query results, execution status, and validation errors in an understandable format.

FR-P5-10 — Result Export

The system shall allow authorized users to export supported result sets, for example as CSV, subject to result-size limits.

FR-P5-11 — Source Synchronization

The system shall support refreshing configured schema information and synchronization status.

FR-P5-12 — Source Selection

The system shall allow users or administrators to select approved sources for supported AI or agent workflows.

FR-P5-13 — Credential Management

The system shall store connector credentials securely and prevent them from being exposed in page responses, ordinary logs, or generated AI context.

Market-data ingestion and management

- FR-P5-14 — Market-Data Provider Integration: The system shall ingest data from configured and authorized market-data providers.
- FR-P5-15 — Instrument Master: The system shall maintain identifiable instruments with their ticker, company name, exchange, currency, asset type, and other available identifying metadata.
- FR-P5-16 — Historical and Intraday Prices: The system shall retrieve supported historical price data, including OHLCV (open, high, low, close, and volume) records for the available frequencies.
- FR-P5-17 — Corporate Actions: The system shall ingest available dividends, stock splits, and other relevant corporate actions.
- FR-P5-18 — Fundamental Data: The system shall retrieve available company fundamentals, valuation metrics, financial statements, and relevant reporting periods.
- FR-P5-19 — Reference and Event Data: The system shall support applicable market indices, sector classifications, trading calendars, and corporate-event information offered by the configured source.
- FR-P5-20 — User Dataset Import: The system shall allow users to import supported CSV or spreadsheet datasets and inspect their schemas before analysis.
- FR-P5-21 — Data Preview: The system shall provide a preview of imported or queried records, including column types, units, date ranges, and missing-value counts where available.
- FR-P5-22 — Data Cleaning: The system shall support controlled cleaning operations, including handling missing values, duplicate records, invalid values, and inconsistent date or numeric formats.
- FR-P5-23 — Historical Backfill: The system shall support retrieving missing historical ranges when the provider permits it and the account's data entitlement allows it.
- FR-P5-24 — Incremental Refresh: The system shall refresh supported datasets incrementally and record the last successful update.

- FR-P5-25 — Data Quality Validation: The system shall check for issues such as missing intervals, duplicate timestamps, invalid OHLC relationships, and unexpected gaps, reporting detected problems without silently concealing them.
- FR-P5-26 — Currency and Adjustment Handling: The system shall preserve the currency and adjustment status of price data and apply the configured split or dividend adjustment policy consistently.
- FR-P5-27 — Source and Freshness Metadata: The system shall record the data source, observation timestamp, ingestion time, frequency, and applicable data-quality status.

Data exploration and analysis

- FR-P5-28 — Structured Query Builder: Users shall be able to filter and query supported financial datasets using fields, conditions, date ranges, and sorting.
- FR-P5-29 — Controlled SQL Interface: The system shall provide an authorized SQL interface for compatible relational datasets, with query validation, read/write permission enforcement, timeouts, and result limits.
- FR-P5-30 — Data Transformations: Users shall be able to apply supported transformations such as calculated columns, percentage returns, rolling statistics, aggregations, and date-based grouping.
- FR-P5-31 — Dataset Joins: Users shall be able to combine compatible datasets using explicit join keys and inspect the resulting schema.
- FR-P5-32 — Saved Queries and Datasets: Users shall be able to save supported queries, query parameters, transformations, and resulting dataset definitions for reuse.
- FR-P5-33 — Analysis Export: The system shall export permitted results to supported formats, including CSV, with units and column definitions preserved where applicable.
- FR-P5-34 — Data Lineage: The system shall identify the input datasets and transformation steps used to produce a saved derived dataset.
- FR-P5-35 — Data Snapshot Identification: The system shall identify the relevant data version or observation range used in a backtest or saved analysis so the user can understand which inputs produced the result.

Page relationships: P5 implements the financial-data foundation. P2 and P3 depend on it for analytics; P4 uses its instrument identifiers; P6 uses its data to identify signals; P7 uses consistent historical data for simulation. P8 configures providers, refresh behavior, and available markets.

Non-Functional Requirements

NFR-P5-01 — Least Privilege: Source connections shall use the minimum database privileges needed for their configured purpose.

NFR-P5-02 — Query Protection: The system shall enforce query timeouts, row limits, and other appropriate resource limits on structured-data queries.

NFR-P5-03 — Query Performance: Ordinary metadata and paginated query requests should target a 95th-percentile response time of 500 milliseconds under the agreed test conditions, excluding slow external database operations.

NFR-P5-04 — Secret Protection: Connector secrets shall be encrypted or held in a suitable secrets manager and masked in the user interface.

NFR-P5-05 — Failure Isolation: An unavailable external data source shall not make unrelated application pages unusable.

NFR-P5-06 — Auditability: Source changes, query execution where required by policy, and permission changes shall be auditable without retaining unnecessary sensitive result contents.

NFR-P5-07 — Data Boundaries: Data returned to an AI model shall comply with source permissions, user permissions, and configured data-handling policies.

Page Relationships

- **Implements:** Structured-data source management and controlled data exploration.
- **Depends on:** Page 1 for permissions and Page 9 for applicable workspace policies.
- **Extends:** Page 3 through approved structured-data retrieval and Page 6 through authorized data-access tools.
- **Related pages:** Page 8 configures compatible AI integrations; Page 7 can record structured-data tool executions.

PAGE 6 — AI Agents and Workflow Management

Purpose: Allow authorized users to configure, validate, and execute repeatable AI workflows that use approved models, knowledge bases, and tools.

Functional Requirements

FR-P6-01 — Create Workflow

The system shall allow users with the appropriate permissions to create an agent or workflow.

FR-P6-02 — Workflow Configuration

The system shall allow authorized users to set a workflow's name, description, instructions, model, input schema, and supported execution parameters.

FR-P6-03 — Tool Selection

The system shall allow users to select from approved tools and integrations instead of allowing unrestricted tool execution.

FR-P6-04 — Knowledge-Base Selection

The system shall allow workflows to use authorized knowledge bases and document collections.

FR-P6-05 — Data-Source Selection

The system shall allow workflows to use only permitted data sources and operations.

FR-P6-06 — Workflow Validation

The system shall validate required fields, permitted tools, model availability, and input configurations before execution.

FR-P6-07 — Test Execution

The system shall allow users to test a workflow with sample inputs before enabling wider use.

FR-P6-08 — Workflow Versioning

The system shall preserve a record of published workflow configurations or versions so that a previous configuration can be reviewed or restored.

FR-P6-09 — Execution Initiation

The system shall allow an authorized user to execute a workflow manually using validated inputs.

FR-P6-10 — Workflow Status

The system shall display a workflow's state, such as draft, enabled, disabled, or execution unavailable, where applicable.

FR-P6-11 — Approval Gates

The system shall support an approval step before executing configured high-impact actions that require human authorization.

FR-P6-12 — Failure Handling

The system shall define retry, timeout, and failure policies for supported workflow steps.

FR-P6-13 — Workflow Ownership

The system shall associate each workflow with an owner and an appropriate access policy.

FR-P6-14 — Workflow Scheduling

If scheduling is enabled, the system shall allow authorized users to configure, pause, and inspect scheduled executions.

The present agent page explains how to build generic workflows, but a trader also needs to express financial conditions, discover candidates, define strategies, and manage signals.

- FR-P6-15 — Stock Screener: The system shall support stock screening using configured price, return, liquidity, valuation, growth, technical-indicator, and other available criteria.
- FR-P6-16 — Custom Screening Rules: Users shall be able to combine multiple screening conditions using supported logical operators.
- FR-P6-17 — Saved Screens: Users shall be able to save, rename, edit, clone, and rerun screening configurations.
- FR-P6-18 — Strategy Definition: Users shall be able to define a strategy with an instrument universe, time interval, entry conditions, exit conditions, and configurable parameters.
- FR-P6-19 — Indicator-Based Rules: Strategies shall be able to use supported calculated indicators, price conditions, and configured financial fields as inputs.
- FR-P6-20 — Position Sizing Rules: Users shall be able to configure supported position-sizing approaches, such as fixed allocation, fixed quantity, or risk-based sizing, subject to the strategy's constraints.
- FR-P6-21 — Risk and Exit Rules: Strategies shall be able to define supported exit conditions and risk controls, such as stop-loss rules, take-profit rules, and maximum position limits.
- FR-P6-22 — Signal Generation: The system shall generate identifiable signals when configured screening or strategy conditions are satisfied.
- FR-P6-23 — Signal Explanation: Each generated signal shall include the rule or conditions that triggered it and the relevant observations.
- FR-P6-24 — Threshold Alerts: Users shall be able to configure alerts triggered by supported conditions, such as price thresholds, percentage movements, indicator values, or screening results.
- FR-P6-25 — Scheduled Scans: Authorized users shall be able to schedule supported screens or research workflows at configured times or frequencies.
- FR-P6-26 — Signal History: The system shall record signal timestamps, relevant inputs, triggering conditions, and subsequent status changes.
- FR-P6-27 — Strategy Versioning: The system shall preserve strategy versions and their configuration differences when a strategy is edited.

- FR-P6-28 — Strategy Variants: Users shall be able to create and compare parameter variants of a strategy.
- FR-P6-29 — AI-Assisted Strategy Drafting: The assistant shall be able to draft a strategy definition from a user description, but the result must be validated against supported inputs and rules before use.
- FR-P6-30 — Strategy Readiness Validation: Before a strategy is sent to the backtesting engine, the system shall validate its required data, rule configuration, instrument eligibility, and parameters.
- FR-P6-31 — Execution Authorization: A strategy that generates signals shall not automatically place real-money trades unless a separately authorized trading workflow explicitly permits it.

Page relationships: P6 extends P3 with reusable financial workflows and depends on P5 for data, P8 for configuration, and P7 for backtesting and execution. P2 presents the resulting screeners, signals, and alerts.

Non-Functional Requirements

NFR-P6-01 — Execution Isolation: Each workflow execution shall have an identifiable run record and shall not unintentionally share mutable execution state with another run.

NFR-P6-02 — Tool Security: Agents shall execute only explicitly authorized tools and operations.

NFR-P6-03 — Bounded Execution: Workflow steps shall have configured timeouts, resource limits, and retry limits.

NFR-P6-04 — Reproducibility: Execution records shall identify the workflow version and available model/configuration versions used for the run.

NFR-P6-05 — Human Oversight: Approval-gated actions shall not execute until the required approval has been recorded.

NFR-P6-06 — Maintainability: New tools and models should be integrated through defined interfaces instead of requiring changes throughout the application.

NFR-P6-07 — Failure Recovery: A failed workflow shall produce an identifiable failure state and sufficient diagnostics for authorized troubleshooting.

Page Relationships

- **Implements:** Agent configuration, workflow validation, and execution initiation.

- **Depends on:** Page 1 for authorization, Page 8 for model settings, Page 4 for knowledge retrieval, and Page 5 for structured-data tools.
- **Extends:** Page 3's model and retrieval capabilities with multi-step execution; approval steps extend workflows that require authorization.
- **Creates records for:** Page 7's execution monitor and Page 2's dashboard metrics.

PAGE 7 — Execution History and Observability

Purpose: Provide an auditable view of conversations, indexing jobs, and agent executions, including their statuses and operational diagnostics.

Functional Requirements

FR-P7-01 — Run Listing

The system shall display authorized executions with identifiers, execution types, start times, durations, and statuses.

FR-P7-02 — Status Tracking

The system shall represent supported states such as queued, running, completed, failed, and cancelled.

FR-P7-03 — Run Filtering

The system shall provide filters for execution type, status, time range, and other available metadata.

FR-P7-04 — Execution Details

The system shall provide detailed information about a selected execution, including its inputs and outputs where authorized.

FR-P7-05 — Step Tracing

The system shall record the sequence and status of relevant execution steps, including retrieval and tool invocation steps when applicable.

FR-P7-06 — Error Diagnostics

The system shall display actionable error information and correlation identifiers for failed executions.

FR-P7-07 — Retry

The system shall allow authorized users to retry eligible failed executions according to the applicable retry policy.

FR-P7-08 — Cancellation

The system shall allow authorized users to cancel eligible running jobs when the relevant worker or provider supports cancellation.

FR-P7-09 — Usage Metrics

The system shall display available duration, token usage, retry counts, and other execution metrics.

FR-P7-10 — Log Search

The system shall allow authorized users to search or filter available execution events.

FR-P7-11 — Export

The system shall support exporting authorized execution metadata or diagnostic records in a defined format.

FR-P7-12 — Audit History

The system shall retain relevant security and administrative audit events according to the configured retention policy.

FR-P7-13 — Completion Notifications

Where notifications are enabled, the system shall notify users about selected execution outcomes.

Backtesting and strategy evaluation

- FR-P7-14 — Backtest Configuration: Users shall be able to configure a backtest using a strategy version, instrument or universe, date range, timeframe, initial capital, and applicable execution settings.
- FR-P7-15 — Transaction Cost Modeling: The backtesting engine shall support configurable commissions, fees, slippage, and other applicable trading costs. Additional taxes or charges shall be included where relevant data and rules are available.
- FR-P7-16 — Backtest Performance Metrics: The system shall calculate supported metrics such as net return, annualized return, volatility, Sharpe ratio, Sortino ratio,

maximum drawdown, win rate, profit factor, trade count, average trade P&L, and exposure.

- FR-P7-17 — Equity Curve: The system shall plot backtested portfolio value or returns over time.
- FR-P7-18 — Benchmark Comparison: The system shall compare backtest results with a selected benchmark or reference strategy over a comparable period.
- FR-P7-19 — Drawdown Analysis: The system shall display drawdown periods, their magnitude, and available recovery information.
- FR-P7-20 — Periodic Return Analysis: The system shall display supported periodic results, such as monthly and annual returns, and highlight periods that materially affect the overall result.
- FR-P7-21 — Trade-Level Results: The system shall provide a trade table with available entry and exit times, prices, direction, duration, costs, P&L, and triggering signals.
- FR-P7-22 — Trade Visualization: The system shall allow a user to inspect supported trades in the relevant historical price chart.
- FR-P7-23 — Backtest Comparison: Users shall be able to compare saved backtests using their strategy versions, assumptions, metrics, and performance charts.
- FR-P7-24 — Parameter Testing: Where supported, users shall be able to evaluate strategy parameter variants and compare their results.
- FR-P7-25 — Out-of-Sample Validation: The system shall support evaluation on separate historical periods or walk-forward procedures when the backtesting engine offers those capabilities.
- FR-P7-26 — Backtest Integrity Checks: The engine shall detect or prevent supported forms of look-ahead bias and shall document how survivorship bias, corporate actions, missing data, and execution assumptions are handled.
- FR-P7-27 — Backtest Reproducibility: Every saved run shall retain the strategy version, data reference, parameter settings, benchmark, and cost assumptions needed to understand or repeat the result.
- FR-P7-28 — Result Export: Users shall be able to export supported summary metrics, charts or chart data, and trade records.

Paper trading and actual execution

The following requirements are applicable when the corresponding trading mode is implemented.

- FR-P7-29 — Paper Portfolio: The system shall support simulated capital, positions, cash balances, and P&L without submitting real orders.
- FR-P7-30 — Simulated Order Management: In paper mode, the system shall support simulated order submission, amendment, cancellation, and lifecycle tracking using the configured simulation rules.
- FR-P7-31 — Trade Blotter: Where the required data exists, the system shall provide a consolidated view of simulated trades and imported or broker-sourced orders.

- FR-P7-32 — Order Status Tracking: Where a broker integration is enabled, the system shall track reported order states such as pending, accepted, partially filled, filled, cancelled, and rejected, according to the provider's status model.
- FR-P7-33 — Broker-Connected Positions: If account access is authorized, the system shall display reported holdings, open positions, cash balances, and order information.
- FR-P7-34 — Real-Order Confirmation: A real-money order shall require an explicit confirmation step showing the instrument, side, quantity, order type, applicable price parameters, and selected broker account.
- FR-P7-35 — Trading Risk Checks: Where real orders are supported, the system shall apply configured pre-trade limits and reject orders that violate required controls.
- FR-P7-36 — Order Reconciliation: The system shall reconcile locally recorded orders and positions with the broker's reported state when broker integration is enabled.
- FR-P7-37 — Trade Journal: Users shall be able to record or annotate trades with the rationale, strategy, timestamps, tags, and post-trade observations.
- FR-P7-38 — Trading Notifications: Where notification delivery is configured, users shall be able to receive supported alerts for strategy completion, simulated fills, order-status changes, and execution errors.

Page relationships: P7 implements backtesting and the permitted execution modes. It depends on P5 for historical and current market data, P6 for strategy definitions, P8 for broker and cost settings, and P1 for authorization. P2 displays summarized results; P3 can explain a selected backtest through an approved analytical workflow.

Non-Functional Requirements

NFR-P7-01 — Observability: Each execution shall have a stable identifier that connects its associated backend logs, steps, and status changes.

NFR-P7-02 — Log Security: Secrets, authentication tokens, and unnecessary sensitive prompt contents shall be redacted from diagnostic output.

NFR-P7-03 — Status Freshness: Running executions should reflect state changes within five seconds under normal operation when the backend can report them.

NFR-P7-04 — Reliability: A transient monitoring or UI error shall not delete a valid execution record.

NFR-P7-05 — Access Control: Detailed inputs, outputs, and logs shall be visible only to users authorized to inspect the related execution.

NFR-P7-06 — Retention: Execution logs, audit events, and detailed traces shall follow separately defined retention policies.

NFR-P7-07 — Scalability: Execution history retrieval shall use pagination and appropriate database indexes to prevent unbounded history queries.

Page Relationships

- **Implements:** Execution monitoring, operational diagnostics, retry controls, and history inspection.
- **Depends on:** Page 1 for access control and execution records created by Pages 3, 4, and 6.
- **Extends:** Page 6 with retry and cancellation actions; Page 3 with conversation-generation diagnostics.
- **Provides data to:** Page 2's dashboard and, where authorized, Page 9's administrative audit views.

PAGE 8 — Model, RAG, and Integration Settings

Purpose: Configure the language model, embedding model, vector database, retrieval pipeline, and supported AI integrations.

Functional Requirements

FR-P8-01 — Ollama Configuration

The system shall allow authorized administrators to configure the Ollama service address and other required connection parameters.

FR-P8-02 — Connection Validation

The system shall provide a way to test whether the configured Ollama service is reachable.

FR-P8-03 — Model Discovery

The system shall display the available models that can be discovered or explicitly configured through the inference service.

FR-P8-04 — Default Model

The system shall allow an authorized administrator to select the default generation model.

FR-P8-05 — Inference Parameters

The system shall allow permitted configuration of supported generation parameters, such as temperature, output-token limit, and sampling parameters.

FR-P8-06 — Embedding Model

The system shall allow an administrator to select the embedding model used for indexing and retrieval.

FR-P8-07 — Vector Database Configuration

The system shall allow authorized administrators to configure and test the supported vector database connection.

FR-P8-08 — Chunking Configuration

The system shall allow configuration of supported chunk sizes, chunk overlap, and text-processing options.

FR-P8-09 — Retrieval Configuration

The system shall allow configuration of retrieval parameters such as top-k results, similarity thresholds, and eligible knowledge-base collections.

FR-P8-10 — Metadata Filtering

The system shall apply appropriate metadata and permission filters during retrieval.

FR-P8-11 — RAG Testing

The system shall provide a test interface for a query that shows retrieved documents or chunks, metadata, similarity results where supported, and relevant configuration information.

FR-P8-12 — Service Diagnostics

The system shall display the connection and readiness status of configured AI dependencies.

FR-P8-13 — Configuration Versioning

The system shall record relevant configuration changes so that administrators can identify which settings were active for an execution.

FR-P8-14 — Configuration Validation

The system shall reject invalid settings and prevent an invalid configuration from being activated without a clear error report.

FR-P8-15 — Model Compatibility

The system shall check or document embedding-model compatibility when an embedding configuration changes and require re-indexing when necessary.

FR-P8-16 — Provider Abstraction

The backend shall expose defined interfaces for inference, embeddings, and retrieval so supported components can be replaced without rewriting unrelated application functionality.

- FR-P8-17 — Market-Data Provider Management: Administrators shall be able to configure supported market-data providers, validate connections, and inspect connection status.
- FR-P8-18 — Data Coverage Inspection: The system shall expose the instruments, historical ranges, frequencies, and data fields available through each configured provider, where that information can be retrieved.
- FR-P8-19 — Market and Exchange Selection: Authorized users shall be able to configure supported markets, exchanges, and instrument universes.
- FR-P8-20 — Data Refresh Configuration: Administrators shall be able to configure supported refresh intervals, historical backfill settings, and applicable provider limits.
- FR-P8-21 — Financial Display Settings: Users shall be able to configure supported display preferences, including base currency, date conventions, timezone, and preferred benchmark.
- FR-P8-22 — Trading Calendar Configuration: The system shall use the appropriate configured exchange calendar and trading sessions for relevant calculations and scheduled tasks.
- FR-P8-23 — Indicator Defaults: The system shall support configuration of default chart indicators, their parameters, and supported calculation conventions.
- FR-P8-24 — Risk Metric Assumptions: Authorized users shall be able to configure applicable assumptions, such as a risk-free rate, benchmark, annualization convention, and risk-measure confidence level.
- FR-P8-25 — Backtesting Cost Assumptions: The system shall allow configuration of applicable commission, slippage, and other supported transaction-cost assumptions.
- FR-P8-26 — Broker Connection Management: If broker integration is implemented, administrators or authorized users shall be able to connect, validate, inspect, and revoke supported broker connections.
- FR-P8-27 — Broker Permission Modes: The system shall distinguish read-only broker access, paper-trading access, and trade-capable access where supported by the integration.
- FR-P8-28 — Trading Safety Controls: The system shall allow authorized administrators to enable or disable trade-capable operations and configure their applicable permission and risk constraints.
- FR-P8-29 — Financial Tool Permissions: Administrators shall be able to control which AI tools can access market data, portfolio data, strategy execution, and trading integrations.

- FR-P8-30 — Provider-Specific Diagnostics: The system shall explain relevant connection, data entitlement, authentication, or rate-limit errors without revealing secrets.
- FR-P8-31 — Configuration Validation: The system shall validate market-data, broker, risk, and model settings before activation and retain the relevant configuration version for subsequent analysis.

Page relationships: P8 implements provider and integration configuration. It extends P3–P7 by making model, RAG, market-data, risk, and optional broker configurations available to those pages. P5 depends on its data-source configuration, while P7 depends on its execution and safety settings.

Non-Functional Requirements

NFR-P8-01 — Configuration Security: Connection secrets shall be stored using an appropriate secret-management mechanism and shall not be displayed in plaintext after saving.

NFR-P8-02 — Change Traceability: Relevant configuration changes shall record the actor, timestamp, affected setting, and outcome.

NFR-P8-03 — Safe Activation: The application shall validate new settings before activation and preserve a recoverable previous configuration where appropriate.

NFR-P8-04 — Inference Isolation: An unavailable model shall produce a controlled error state without crashing the application backend.

NFR-P8-05 — Retrieval Consistency: Indexing and query-time retrieval shall use compatible embedding dimensions, model versions, and vector-index configurations.

NFR-P8-06 — Extensibility: Model or vector-database integrations shall use modular interfaces and documented configuration contracts.

NFR-P8-07 — Diagnostic Security: Service diagnostics shall reveal sufficient error information for troubleshooting without disclosing secrets.

NFR-P8-08 — Performance Configuration: Changes to generation or retrieval parameters shall be validated against configured limits to avoid uncontrolled resource consumption.

Page Relationships

- **Implements:** AI service configuration, model selection, RAG tuning, and integration diagnostics.
- **Depends on:** Page 1 for administrative authorization and the relevant service adapters for Ollama, PostgreSQL, and the vector database.

- **Extends:** Page 3 with inference and retrieval controls; Page 4 with chunking and embedding controls; Page 6 with agent execution settings.
- **Affects:** Pages 2, 3, 4, 5, 6, and 7 through service availability and execution configuration.

PAGE 9 — Account, Workspace, and Preferences

Purpose: Allow users to manage their personal profile and preferences and allow authorized administrators to manage workspace settings.

Functional Requirements

FR-P9-01 — View Profile

The system shall allow users to view their account profile and assigned role.

FR-P9-02 — Update Profile

The system shall allow users to update permitted profile fields.

FR-P9-03 — Security Preferences

The system shall allow users to manage supported account-security settings, including password changes when password authentication is enabled.

FR-P9-04 — Interface Preferences

The system shall support configured preferences such as theme, language, timezone, and other user-interface options.

FR-P9-05 — Notification Preferences

The system shall allow users to configure supported notifications.

FR-P9-06 — Workspace Settings

Authorized administrators shall be able to view and modify permitted workspace-level settings.

FR-P9-07 — Member Management

Authorized administrators shall be able to invite, deactivate, or remove workspace members according to the defined access policy.

FR-P9-08 — Role Management

Authorized administrators shall be able to assign supported roles, subject to safeguards that prevent unauthorized privilege escalation or removal of the last required administrator.

FR-P9-09 — API Credentials

If API-key functionality is enabled, the system shall allow authorized users to create, inspect metadata for, revoke, and rotate scoped API credentials.

FR-P9-10 — Data Export

The system shall support exporting a user's authorized application data through a documented export process.

FR-P9-11 — Account Deletion

The system shall support account deletion or deactivation according to the application's identity, retention, and ownership policies.

FR-P9-12 — Privacy and Retention

Authorized administrators shall be able to inspect and manage applicable data-retention settings.

FR-P9-13 — Administrative Audit

Authorized administrators shall be able to inspect relevant administrative and security audit records.

FR-P9-14 — Confirmation for Sensitive Actions

The system shall require appropriate confirmation for sensitive actions such as account deletion, API-key revocation, and member removal.

- FR-P9-15 — Saved Dashboards: Users shall be able to save, rename, duplicate, and restore dashboard layouts.
- FR-P9-16 — Saved Watchlists: Users shall be able to manage their watchlists and their display preferences through their workspace.
- FR-P9-17 — Saved Screeners and Queries: Users shall be able to manage their saved stock screens, data queries, transformations, and supported dataset definitions.
- FR-P9-18 — Saved Strategies: Users shall be able to access and organize their strategy definitions, versions, and related backtest records.

- FR-P9-19 — Saved Reports: Users shall be able to save, reopen, export, and organize supported analytical reports.
- FR-P9-20 — Research Organization: Users shall be able to organize permitted research documents, AI conversations, and financial analyses into collections or projects.
- FR-P9-21 — Personal Market Preferences: Users shall be able to set their default market, preferred watchlist, benchmark, chart preferences, and supported display currency.
- FR-P9-22 — Alert Preferences: Users shall be able to configure supported notification channels and preferences for watchlist, price, signal, and workflow alerts.
- FR-P9-23 — Workspace Sharing: If sharing is enabled, users shall be able to share supported dashboards, datasets, research collections, or reports with specific authorized workspace members.
- FR-P9-24 — Saved-Item Access Control: The system shall enforce the required ownership and sharing permissions for saved analyses, strategies, datasets, and reports.
- FR-P9-25 — Personal Data Export and Deletion: Users shall be able to export or request deletion of eligible personal workspace data according to the documented retention and ownership policies.

Page relationships: P9 implements saved workspace and personalization features. It depends on P1 for authorization and extends P2, P3, P5, P6, and P7 with persistence, organization, and reuse of user-created work.

Non-Functional Requirements

NFR-P9-01 — Privacy: Users shall not be able to view or alter another user's private account information without explicit authorization.

NFR-P9-02 — Sensitive Changes: Privilege changes, API-key operations, account deletion, and security-setting changes shall be authorized and auditable.

NFR-P9-03 — Secret Handling: The interface shall never redisplay stored API secrets in plaintext after the creation flow.

NFR-P9-04 — Preference Persistence: Saved preferences shall persist across sessions and supported devices or browsers according to the account model.

NFR-P9-05 — Change Confirmation: Destructive changes shall communicate their consequences and require the defined confirmation step.

NFR-P9-06 — Administrative Isolation: Workspace administration functionality shall be unavailable to users without the relevant privileges.

NFR-P9-07 — Data Lifecycle: Account deletion and export shall follow documented rules for owned records, shared records, audit history, backups, and legal or operational retention constraints.

Page Relationships

- **Implements:** Profile, account, workspace, preference, member-management, and retention interfaces.
- **Depends on:** Page 1 for authentication and authorization, and PostgreSQL for applicable user and workspace metadata.
- **Extends:** Page 1 with user and security management and Page 7 with administrative audit access.
- **Affects:** Every page through workspace membership, access permissions, personalization, and retention policies.

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3. Cross-Cutting Non-Functional Requirements

The following requirements apply to the full application, not just one screen. They must be implemented and tested across all nine pages.

Security and Privacy

NFR-G-01 — Authorization Enforcement

Every protected backend endpoint shall enforce authentication and permission checks. Hiding a frontend control shall not count as authorization.

NFR-G-02 — Data Encryption

Sensitive information shall be protected in transit using TLS in deployed environments and encrypted at rest where required by the deployment threat model.

NFR-G-03 — Secrets Management

Database credentials, API keys, and other secrets shall be held outside source code and ordinary application configuration files committed to version control.

NFR-G-04 — Input Validation

All user inputs, uploaded files, query parameters, and configuration changes shall be validated at the backend.

NFR-G-05 — Injection Resistance

Database operations shall use parameterized queries or safe query builders wherever applicable. Other interpreter and command-execution boundaries shall be protected against injection.

NFR-G-06 — Tenant and Ownership Isolation

Where the application supports multiple users or workspaces, records shall be scoped appropriately, and access checks shall be enforced consistently across the application.

NFR-G-07 — Sensitive Logging

Ordinary application logs shall not expose passwords, tokens, database credentials, or unnecessarily retained private content.

NFR-G-08 — Auditability

Administrative and security-sensitive actions shall produce timestamped audit records with the actor, action, affected resource, and outcome where applicable.

Performance and Capacity

NFR-G-09 — API Latency

Ordinary backend operations that do not invoke a model, wait on a long-running job, or query a slow external source should target a 95th-percentile response time of 500 milliseconds under an agreed reference workload.

NFR-G-10 — Long-Running Tasks

Document indexing, embeddings, and multi-step agent executions shall be processed through suitable asynchronous workers or equivalent mechanisms.

NFR-G-11 — Bounded Resource Usage

The system shall define configurable request sizes, upload limits, generation limits, query limits, execution timeouts, and retry budgets.

NFR-G-12 — Capacity Testing

Performance targets shall be verified using a documented reference machine, dataset, model, and workload. Local model performance shall not be evaluated without recording the relevant hardware and model configuration.

Reliability and Data Integrity

NFR-G-13 — Controlled Failure

If Ollama, PostgreSQL, or the vector database becomes unavailable, the application shall show an understandable failure state and preserve unrelated services and persisted data where possible.

NFR-G-14 — Data Consistency

The system shall define recovery and reconciliation procedures for operations that write to both PostgreSQL and the vector database, since they generally cannot participate in one ordinary cross-database transaction.

NFR-G-15 — Backup and Restore

The application shall have documented backup and restoration procedures for PostgreSQL and the configuration or metadata necessary to rebuild vector indexes.

NFR-G-16 — Recovery Targets

The deployment owner shall define and test a recovery time objective (RTO) and recovery point objective (RPO) that match the application's criticality.

NFR-G-17 — Idempotent Operations

Retryable background tasks shall be designed to avoid duplicate active records, duplicated indexing, or repeated side effects where applicable.

Usability and Accessibility

NFR-G-18 — Consistent Interaction

All pages shall provide consistent navigation, loading indicators, error feedback, confirmation behavior, and accessible form controls.

NFR-G-19 — Accessibility

The interface shall target WCAG 2.2 Level AA, including keyboard accessibility, meaningful labels, adequate contrast, and accessible status announcements.

NFR-G-20 — Responsive Design

All nine pages shall remain usable across the declared supported screen sizes and browsers.

NFR-G-21 — Error Understandability

Users shall receive actionable messages without being exposed to raw stack traces, internal secrets, or unnecessary implementation details.

Maintainability and Compatibility

NFR-G-22 — Modular Architecture

Frontend pages, backend business logic, storage adapters, inference adapters, embedding services, and retrieval logic shall have clearly defined responsibilities.

NFR-G-23 — Automated Testing

The project shall include unit tests for important business logic, integration tests for database and AI-service interactions, and end-to-end tests for core user flows.

NFR-G-24 — API Documentation

Backend endpoints and their authentication, request schemas, response formats, errors, and permissions shall be documented.

NFR-G-25 — Configuration Separation

Environment-specific configuration shall be separated from application code.

NFR-G-26 — Database Migrations

Database schema changes shall be versioned and applied through controlled migration procedures.

NFR-G-27 — Dependency Management

Application dependencies shall be versioned, and dependency updates shall be tested before deployment.

AI Quality and Traceability

NFR-G-28 — Model Identification

The system shall record the model and relevant configuration version for each AI operation whenever the information is available.

NFR-G-29 — Retrieval Traceability

RAG executions shall retain sufficient metadata to identify the knowledge-base version and source chunks used for retrieval.

NFR-G-30 — No Unsupported Guarantees

The system shall not represent generated output as verified fact solely because a language model produced it. Where appropriate, the UI shall explain uncertainty or insufficient evidence.

NFR-G-31 — Retrieval Permission Enforcement

Permission filtering shall be applied before private retrieved content is provided to a model, not merely after the response is generated.

NFR-G-32 — Model and Index Compatibility

The system shall prevent incompatible embedding models or vector dimensions from being used against an existing index without the required migration or re-indexing.

NFR-G-33 — Model Replacement

The model-integration layer should allow an authorized administrator to replace the local model or add another supported provider without changing unrelated frontend pages.

Monitoring and Operations

NFR-G-34 — Health Checks

The system shall expose or internally maintain health checks for critical components.

NFR-G-35 — Structured Logs

Logs shall support correlation between API requests, document-processing jobs, model calls, and workflow executions.

NFR-G-36 — Operational Monitoring

The system shall collect available operational information such as error rates, response latency, background-job failures, and resource usage.

NFR-G-37 — Graceful Shutdown

The application shall stop accepting new work and handle in-flight work according to its shutdown policy when a backend service or worker stops.

NFR-G-38 — Deployment Reproducibility

The system shall document how to configure, start, stop, and troubleshoot the application and its required dependencies.

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4. Cross-Page Dependency and Extension Matrix

Page	Implements	Extends	Depends on / Includes
P1 — Authentication	Sign-in, sessions, access checks	Password recovery and additional authentication factors, when enabled	Identity mechanism and user records
P2 — Dashboard	Overview, service health, quick actions	Navigation to recent work and detailed views	P1, P3, P4, P7, P8
P3 — AI Assistant	Conversations and model responses	RAG, structured-data retrieval, feedback, export	P1, P4, P5, P8
P4 — Knowledge Base	Upload, extraction, chunking, indexing	Semantic retrieval for chat and agents	P1, P8, PostgreSQL, vector database
P5 — Data Sources	Connection management and structured-data access	Approved data retrieval for AI and agents	P1, P9 and configured data sources
P6 — Agents and Workflows	Workflow configuration and execution	Multi-step tasks and conditional approval	P1, P4, P5, P8
P7 — Execution Monitor	Run history, logs, diagnostics	Retry, cancellation, and completion notifications	P1, P3, P4, P6
P8 — Model/RAG Settings	Models, embeddings, retrieval and connection settings	Advanced retrieval or model configurations	P1, model and storage adapters
P9 — Account/Workspace	Profiles, preferences, members and retention	Administrative and security management	P1, P7 and PostgreSQL

5. Key End-to-End Workflows

Workflow A — RAG-Enabled Question Answering

1. The user signs in through P1.
2. The user opens P3 and submits a question.
3. The backend reads the active inference and retrieval settings from P8.
4. The retrieval service searches the vector database using the configured embedding model.
5. PostgreSQL supplies document metadata and access-control information.
6. Unauthorized results are removed before context is passed to the model.
7. The selected local model generates a response using the retrieved context.
8. P3 displays the answer and source references.
9. Execution metadata is saved for P7, and available activity metrics appear on P2.

Workflow B — Document Ingestion

1. The user uploads a document through P4.
2. The backend validates the file and stores its metadata in PostgreSQL.
3. The extraction and chunking services prepare the document.
4. The configured embedding model generates chunk vectors.
5. The vectors are stored in the vector database with stable document and chunk identifiers.
6. PostgreSQL and the vector database are reconciled if processing fails partway through.
7. P4 displays the final status, P7 exposes supported processing diagnostics, and P2 updates relevant counts.

Workflow C — Agent Execution

1. The user creates or selects a workflow in P6.
2. The system validates its instructions, model, tools, and data-source permissions.
3. The backend reads the applicable configuration from P8.
4. The workflow retrieves authorized knowledge from P4 or structured data from P5.
5. The system executes the workflow within its configured limits and approval requirements.
6. P7 records and displays the execution status, step history, and errors.
7. P2 updates its operational overview when the relevant records become available.

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6. Data Integrity Rules That Must Not Be Omitted

The application must explicitly enforce the following rules because its relational and vector stores have different responsibilities.

DIR-01 — Canonical Metadata: PostgreSQL shall be the authoritative source for document identity, ownership, permissions, lifecycle status, and other structured metadata.

DIR-02 — Vector References: Each indexed chunk shall carry a stable identifier that can be resolved to its document, version, and authorized metadata.

DIR-03 — Authorized Retrieval: The retrieval service shall enforce document and workspace access rules before supplying content to the LLM.

DIR-04 — Indexing State: A document shall not be marked fully indexed until all required processing and indexing steps have succeeded.

DIR-05 — Update Safety: Re-indexing shall distinguish the active document version from any stale vectors and shall avoid exposing a partially updated index as complete.

DIR-06 — Deletion Propagation: A document deletion shall trigger removal or invalidation of corresponding vector entries and shall be recoverable after a partial failure.

DIR-07 — Retry Safety: Background jobs shall use stable operation identifiers or equivalent idempotency controls.

DIR-08 — Configuration Compatibility: A change to the embedding model or vector dimensions shall trigger an explicit compatibility check and, when necessary, a controlled re-indexing process.

DIR-09 — Audit Separation: Operational logs shall not replace the canonical application data or required security audit trail.

DIR-10 — Backup Recovery: Recovery procedures shall restore relational metadata consistently and either restore the corresponding vector data or rebuild the vector index before declaring the knowledge base ready.

11. Additional financial non-functional requirements

Your existing SRS already contains broad security, performance, reliability, and maintainability requirements. Keep those, but add these finance-specific requirements because generic software requirements do not adequately cover market-data correctness or trading safety.

ID	Additional non-functional requirement
NFR-FIN-01 — Data Freshness	The application shall distinguish real-time, delayed, and end-of-day data, displaying the applicable observation timestamp. It shall not label data as real-time unless the configured source and entitlement support that claim.
NFR-FIN-02 — Time and Calendar Accuracy	Financial observations shall use explicit timestamps, consistent timezone conversion, and the appropriate exchange calendar.
NFR-FIN-03 — Numerical Accuracy	Monetary amounts, quantities, returns, and percentage calculations shall use appropriate precision and rounding rules; premature rounding shall not distort analytical results.
NFR-FIN-04 — Methodological Transparency	Financial metrics shall have documented definitions, input conventions, and calculation methods.
NFR-FIN-05 — Corporate-Action Consistency	Price series shall identify their adjustment status, and calculations shall consistently handle configured split and dividend adjustment policies.
NFR-FIN-06 — Reproducibility	A saved analysis or backtest shall identify its relevant data snapshot or version, strategy version, parameters, and applicable assumptions.

NFR-FIN-07 — Backtest Integrity	The backtesting engine shall document the treatment of look-ahead bias, survivorship bias, transaction costs, missing observations, and execution assumptions.
NFR-FIN-08 — Risk-Metric Consistency	Portfolio and strategy risk measures shall use documented return frequencies, benchmark conventions, annualization rules, and other applicable calculation assumptions.
NFR-FIN-09 — Trading Fail-Safe	When real-money execution is enabled, failed authorization, unavailable required risk checks, or unresolved order-state uncertainty shall not cause the system to silently proceed with a new order.
NFR-FIN-10 — Execution Separation	Market-data access, analysis, paper trading, and real-money trading shall have separate permission scopes wherever supported.
NFR-FIN-11 — Audit Trail	Relevant strategy changes, configuration changes, approval decisions, and broker order actions shall be auditable with timestamps and identifiable actors.
NFR-FIN-12 — Data Provenance	Market-data values and financial research used in consequential outputs shall be traceable to their source and applicable observation or publication time where available.
NFR-FIN-13 — Calculation and AI Separation	Deterministic financial calculations shall be performed by validated tools or services rather than relying on free-form language-model arithmetic. AI-generated interpretation shall be distinguishable from computed results.
NFR-FIN-14 — Provider Failure Handling	When a provider becomes unavailable or returns incomplete data, the system shall indicate the issue and avoid presenting stale data as current.

NFR-FIN-15 — Data Licensing	Market-data ingestion, storage, display, export, and sharing shall comply with the configured provider's applicable entitlements and licensing terms.
NFR-FIN-16 — Alert Reliability	Alerts shall include their triggering condition and timestamp and shall support appropriate duplicate-suppression and failure-reporting behavior.
NFR-FIN-17 — Scalable Analytics	Large historical queries, dataset transformations, and backtests shall use bounded queries or background processing where appropriate, without blocking unrelated user interactions.
NFR-FIN-18 — Integration Security	Broker tokens and market-data credentials shall be protected in storage, masked in the interface, and excluded from ordinary diagnostic logs.
NFR-FIN-19 — Graceful Degradation	Market-data, model, or broker-service failure shall not make unrelated pages unusable; affected features shall clearly disclose reduced functionality.
NFR-FIN-20 — Verification	Financial calculations, adjustment policies, portfolio metrics, and backtesting behavior shall be verified with documented reference cases and integration tests.

12. Final cross-page relationship matrix

This matrix consolidates how the additional requirements fit the original nine-page structure.

Page	Implements	Extends	Main dependencies

P1	Authentication and financial-operation permissions	Sensitive-action security	Identity provider, PostgreSQL
P2	Market and portfolio analytics	AI explanations, signal summaries	P1, P3, P5, P6, P7, P8, P9
P3	AI financial analysis	Retrieval, data querying, strategy and report workflows	P1, P4, P5, P6, P7, P8
P4	Financial research and RAG knowledge	Evidence-backed AI responses	P1, P5, P8
P5	Market data and financial-data exploration	Charting, screening, strategy evaluation	P1, P8 and configured data sources
P6	Screeners, signals and strategies	Scheduled research and strategy testing	P1, P3, P5, P7, P8, P9
P7	Backtesting and permitted trading operations	Strategy execution, detailed portfolio and trade analysis	P1, P5, P6, P8, P9
P8	Model, market-data and optional broker settings	Features on P2–P7	P1, configured services
P9	Saved research, preferences, workspace organization	Reuse and sharing across other pages	P1, P2, P3, P5, P6, P7

13. What is essential for the first release?

I would prioritize the implementation so the app becomes a functional financial-analysis product before adding live trading.

Priority 1

Build the analytical foundation

Market data ingestion, data-quality checks, instrument search, interactive charts, watchlists, returns and fundamental metrics, dashboard filters, data freshness indicators, and exports.

Priority 2

Build the AI analyst and research workflow

RAG over financial reports, source-cited answers, natural-language screening, deterministic financial calculations, saved analyses, and source-backed company comparisons.

Priority 3

Build strategy research

Strategy rules, technical signals, configurable alerts, backtesting, realistic trading costs, benchmark comparisons, risk analytics, and trade-level reports.

Priority 4

Add simulation and optional broker execution

Paper portfolios, simulated orders, broker integration, trade-capable permissions, pre-trade risk checks, and explicit real-order confirmation.

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7. Scope Boundary: Fine-Tuning

Fine-tuning is **not required for the initial implementation** of this baseline. The first version can use a supported local model through Ollama, with RAG supplying application-specific knowledge.

To preserve a future extension path, the system should:

- Keep model inference behind a provider interface.
- Record the model name and relevant configuration version with executions.
- Keep embedding generation independent of the chat-completion service.
- Allow the model configuration to be changed without redesigning the nine pages.
- Provide a later, separate model-training workflow if fine-tuning is introduced.

Adding fine-tuning later may require dataset preparation, model-training jobs, evaluation, model versioning, artifact storage, and model deployment controls. It should not be treated as an existing functional capability until these are implemented.

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8. Requirements Verification Checklist

Before considering the application ready for implementation review, verify all of the following.

- Every page has a defined purpose and explicit functional requirements.
- Every page has associated non-functional requirements.
- Authentication and authorization are enforced on the backend.
- PostgreSQL is used for relational records and authoritative metadata.
- The vector database stores embeddings and supports semantic retrieval.
- RAG retrieval applies permissions before model-context construction.
- Document ingestion, re-indexing, and deletion handle partial failures.
- Ollama connectivity and model availability are visible and testable.
- Agents can use only approved tools and data sources.
- Execution records, logs, and diagnostics are accessible to authorized users.
- Account and workspace operations respect permission and retention policies.
- Performance and reliability targets are verified against documented test conditions.
- Backup, restore, logging, privacy, accessibility, and security are included in testing.
- Each cross-page dependency is represented in the architecture and tested end to end.
- Every requirement ID maps to at least one implementation component and one verification method.

Completion criterion: A requirement is not complete merely because its page exists. Its backend behavior, data handling, authorization, error handling, and applicable non-functional constraints must also be implemented and verified.